

CS5340 – Uncertainty Modeling in AI
(Semester 1 : AY2023/24)

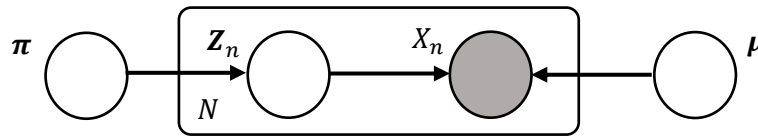
Time Allowed : 2 Hours

INSTRUCTIONS TO STUDENTS

1. Please write your Student Number only. Do not write your name.
2. This assessment paper contains **FOUR** questions and comprises **FIVE** printed pages.
3. Students are required to answer **ALL** questions.
4. Students should write the answers for each question on a new page.
5. This is a **CLOSED BOOK** assessment.
6. Students can bring in **ONE A4 summary sheet**.
7. Non-programmable electronic calculators are allowed.

Question 1

Figure 1.1 shows the Bayesian network of a mixture-of-Bernoulli distribution. $\mathbf{Z} = [z_1 \dots z_K]^T$ is a binary K -dimensional one-hot vector that follows the Categorical distribution parameterized by $\boldsymbol{\pi} = \{\pi_1, \dots, \pi_K\}$, where $\sum_k \pi_k = 1$. X is a binary random variable which follows the k -th Bernoulli distribution indicated by $\mathbf{Z}$. The parameters of the K Bernoulli distributions are $\boldsymbol{\mu} = \{\mu_1, \dots, \mu_K\}$. We are given N *i.i.d* observations of X .

**Figure 1.1**

- i) Ignoring the prior distributions of the parameters $\boldsymbol{\pi}$ and $\boldsymbol{\mu}$, write the **complete log-likelihood** of the given Bayesian network. (5 marks)
- ii) Applying the Expectation Maximization algorithm on the Bayesian network, derive the expression of the **responsibility function** in terms of all the random variables and parameters. (5 marks)
- iii) Derive the expressions for the parameters $\boldsymbol{\pi}$ and $\boldsymbol{\mu}$ after the maximization step. (15 marks)

Show all your workings clearly.

Useful Equations:

1. **Bernoulli distribution:** $p(y | \lambda) = \lambda^y (1 - \lambda)^{1-y}$, where y is a binary random variable and the parameter is subjected to $0 \leq \lambda \leq 1$.
2. **Categorical distribution:** $p(\mathbf{y} | \boldsymbol{\gamma}) = \prod_k \gamma_k^{y_k}$, where the random variable $\mathbf{y} = [y_1 \dots y_K]^T$ is a one-hot vector of K categories, and the parameter is $\boldsymbol{\gamma} = [\gamma_1 \dots \gamma_K]^T$ such that $\sum_k \gamma_k = 1$.

Question 2

Given a dataset $\mathcal{D} = \{x_1, \dots, x_N\}$ of observed values of a random variable X , which are assumed to be drawn independently from a univariate Gaussian distribution: $p(x | \mu, \tau) = \mathcal{N}(x | \mu, \tau^{-1})$ parameterized by the mean μ and precision τ . The prior distributions for μ and τ are given by a univariate Gaussian distribution: $p(\mu | \tau) = \mathcal{N}(\mu | \mu_0, (\lambda_0 \tau)^{-1})$ and a Gamma distribution: $p(\tau) = \text{Gam}(\tau | a_0, b_0)$, respectively. $\{\mu_0, \lambda_0, a_0, b_0\}$ are hyperparameters of the prior distributions.

- i) Using **variation inference** and **mean field approximation**, derive the distributions $q(\mu)$ and $q(\tau)$ that best approximate the conditional distributions $p(\mu | \mathcal{D})$ and $p(\tau | \mathcal{D})$.
(15 marks)
- ii) Derive the expressions for the expected mean μ and precision τ under the approximated distributions $q(\mu)$ and $q(\tau)$.
(2 marks)
- iii) Derive the expressions for the mean μ and precision τ that **maximize the log-likelihood**.
(6 marks)
- iv) Comment on the difference between the mean μ and precision τ obtained from variational inference and maximum log-likelihood.
(2 marks)

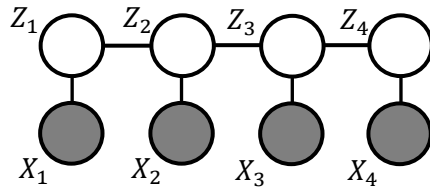
Show all your workings clearly.

Useful Equations:

1. **Gamma distribution:** $\text{Gam}(\tau | a, b) = \frac{1}{\Gamma(a)} b^a \tau^{a-1} \exp(-b\tau)$, where $\Gamma(a)$ is the Gamma function;
2. **Gaussian distribution:** $p(x | \mu, \tau^{-1}) = \left(\frac{\tau}{2\pi}\right)^{1/2} \exp\left\{-\frac{\tau}{2}(x - \mu)^2\right\}$;
3. **Completing the square:** $ax^2 + bx + c = 0 \Rightarrow a(x + d)^2 + e = 0$, where $d = \frac{b}{2a}$ and $e = c - \frac{b^2}{4a}$.

Question 3

Figure 3.1 shows a hidden Markov model with binary latent random variables Z_1, Z_2, Z_3, Z_4 and binary observed random variables $X_1 = 1, X_2 = 0, X_3 = 1, X_4 = 0$. The unary potentials $U_i(Z_i)$ between $Z_i - X_i$ and pairwise potentials $P_{ij}(Z_i, Z_j)$ between $Z_i - Z_j$ are given in Table 3.1.

**Figure 3.1****Unary Potentials**

$U_1(0):$	20	$U_1(1):$	5
$U_2(0):$	5	$U_2(1):$	20
$U_3(0):$	20	$U_3(1):$	5
$U_4(0):$	5	$U_4(1):$	20

Binary Potentials

$P_{12}(0,1):$	10
$P_{23}(0,1):$	5
$P_{34}(0,1):$	10
$P_{12}(1,0):$	5
$P_{23}(1,0):$	10
$P_{34}(1,0):$	5

Table 3.1

- i) Using the **Graph-Cut algorithm**, find the states of the latent random variables Z_1, Z_2, Z_3, Z_4 that maximize the joint probability distribution.

(15 marks)

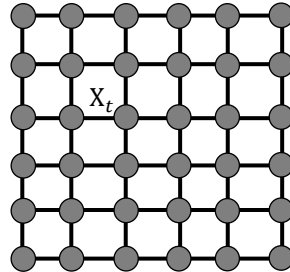
- ii) List all the possible cuts and the respective capacity of each cut.

(10 marks)

Show all your workings clearly.

Question 4

- i) Figure 4.1 shows the Markov Random Field (MRF) of the binary image denoising problem, where each node represents a binary latent random variable X_t of the clean image. The edge potential is given by the Ising model: $\psi(X_s, X_t) = \exp(J X_s X_t)$, where $X_t \in \{-1, +1\}$.

**Figure 4.1**

Given that each pixel of the observed noisy image $Y_t \in \{-1, +1\}$ is related to the respective latent random variable X_t with a directed edge, i.e., $X_t \rightarrow Y_t$, where the likelihood is a Gaussian distribution $p(Y | X) = \mathcal{N}(Y_t | X_t, \sigma^2)$. Both J and σ^2 are fixed hyperparameters.

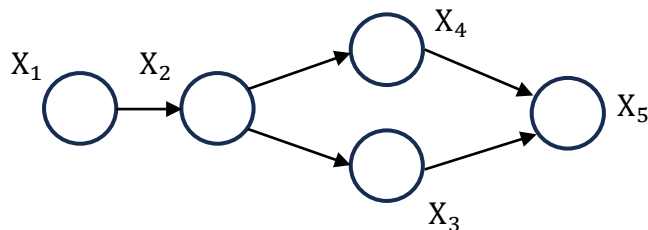
- a) Derive the expression of the full conditional to draw a sample of X_t with **Gibbs sampling**.

(10 marks)

- b) Draw a sample of X_t given that all the other latent random variables are $+1$, $Y_t = +1$ and a random number in $[0,1]$ drawn from a uniform distribution is 0.56. We further take $J = 0.1$ and $\sigma^2 = 0.5$.

(12 marks)

- ii) What is the tree-width of the Bayesian network shown in Figure 4.2? Explain your answer clearly.

(3 marks)**Figure 4.2**

Show all your workings clearly.

Useful Equations:

1. **Gaussian distribution:** $\mathcal{N}(x | \mu, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}$;
2. **Sigmoid function:** $\text{Sigm}(x) = \frac{1}{1 + \exp\{-x\}}$

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